Ruby Shrestha

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Research Interests

Reliable Deep Generative Model; Causal Generative Modeling; Controllable Generation; Robustness to Distribution Shift; Trustworthy ML

Education

PhD, Computing and Information Sciences 08/2022–Present
Rochester Institute of Technology (RIT), Rochester, NY, USA GPA: 4.00/4.00
Advisor: Dr. Linwei Wang
Relevant Courses: Quantitative Foundations, Deep Learning, Statistical Machine Learning, Non-convex Optimization, Data Science & Analytics, Visual Analytics

B.Sc., Computer Science and Information Technology 10/2013–04/2018
Tribhuvan University, Nepal 84.14%

Technical Skills

Programming:	Python, SQL
Machine Learning:	PyTorch, Scikit-learn, NumPy, Pandas
Deep Learning / Research:	Diffusion Models, Generative Models, Causal Modeling, Interventional and Counterfactual Generation
Experiment Tracking:	Weights & Biases, MLflow
Data Systems:	MySQL, Elasticsearch
Visualization:	Tableau, Looker Studio
Tools / Platforms:	Git, Linux, Mac OS, Windows

Research Experience

Graduate Research Assistant 2022–Present
Computational Biomedicine Lab, Rochester Institute of Technology, Rochester, NY, USA

- Conducted a systematic study of how attribute correlations in training data impair generative models' ability to produce underrepresented attribute combinations, and the extent to which existing mitigation strategies address this impact.
- Developed a deep generative modeling method for robust image generation under correlated attributes in the training distribution, leading to a first-author publication at ICLR 2026.
- Researches on causal generative models for controllable and counterfactual image generation, aiming for accurate generation under varying strengths of attribute relationships.

Independent Research Study as Alumna 05/2018–08/2018
Deerwalk Institute of Technology, Tribhuvan University, Nepal

- Empirically assessed a mixed-model path-finding approach combining irregularities-tolerant dead-end filling with A* search for convoluted maze paths.

Research Assistant 03/2017–05/2017
Deerwalk Institute of Technology, Tribhuvan University, Nepal

- Studied student perspectives on blended learning pedagogy at Deerwalk Institute of Technology.

Publications

ProReGen: Progressive Residual Generation under Attribute Correlations
R. Shrestha, A. Gopi, C. Meisenzahl, B. Lekhak, L. Wang
International Conference on Learning Representations (ICLR), 2026. Conference Article.

DATS: Difficulty-Aware Task Sampler for Meta-Learning Physics-Informed Neural Networks
M. Toloubidokhti, Y. Ye, R. Missel, X. Jiang, N. Kumar, **R. Shrestha**, L. Wang
International Conference on Learning Representations (ICLR), 2024. Conference Article.

Distributionally Robust Optimization and Invariant Representation Learning for Addressing Subgroup Underrepresentation: Mechanisms and Limitations
N. Kumar*, **R. Shrestha***, Z. Li, L. Wang *Equal contribution
Fairness of AI in Medical Imaging (FAIMI @ MICCAI), 2023. Workshop Article.

Maze Solving with Anomaly Tolerant Dead End Filling and A*: Implementation and Analysis
R. Shrestha
Deerwalk Journal of Computer Science and Information Technology, 2019. Journal Article.

Industry Experience

Software Consultant 04/2021–07/2022
Saugal Trading Company Pvt. Ltd., Nepal

Senior Software Engineer, Development Manager 01/2021–03/2021
Software Engineer, Team Lead 04/2019–01/2021
Software Engineer 05/2018–03/2019
Software Engineering Intern 09/2015–04/2018
Deerwalk Inc. / Cedar Gate Technologies Company

- Worked on and led projects to optimize Elasticsearch queries and extract healthcare data from Elasticsearch cluster for creating descriptive, prescriptive, and trend-based, industry-standard healthcare reports and data visualizations.

Honors & Awards

- Dr. Russell & Melissa Bessette Award for Doctoral Student Excellence, RIT (2023)
- National Female Education Award, Government of Nepal (2019)
- Winner, International Girls in ICT Day Hackathon (2016)
- Outstanding Cambridge Learners Award – Top in Country (2013)

Poster Presentations

ProReGen: Progressive Residual Generation under Attribute Correlations
The Fourteenth International Conference on Learning Representations, 2026

Conference Presentations

Maze Solving with Anomaly Tolerant Dead End Filling and A*: Implementation and Analysis
The Fifth International Conference on Information Technology for Development, 2018

Blended Learning in DWIT College: From Learners Perspective
The Fourth International Conference on Information Technology for Development, 2017

Teaching Experience

Workshop Instructor 01/2019–03/2019
Deerwalk Institute of Technology, Tribhuvan University, Nepal

- Instructed a workshop on fundamental concepts of data management and analysis with Elasticsearch.

Guest Trainer 07/2018
Local Development Training Academy, Nepal

- Delivered a three-day hands-on training program on cloud services, content management systems, and project management tools.

Guest Trainer 07/2016
Local Development Training Academy, Nepal

- Delivered a three-hour interactive training session on mobile application usage, security, and earning.